

In-depth experimental study of pyrolysis characteristics of raw and cooking treated shrimp shell samples

Pengchao Zhang^a, Hongyun Hu^{a,*}, Hua Tang^a, Yuhan Yang^a, Huan Liu^a, Qiang Lu^b, Xian Li^a, Nakorn Worasuwanarak^c, Hong Yao^a

^a State Key Laboratory of Coal Combustion, School of Energy and Power Engineering, Huazhong University of Science and Technology, Wuhan 430074, China

^b National Engineering Laboratory for Biomass Power Generation Equipment, North China Electric Power University, Beijing 102206, China

^c The Joint Graduate School of Energy and Environment, Center of Excellence on Energy Technology and Environment, King Mongkut's University of Technology Thonburi, 126 Pracha-Uthit Rd., Bangmod, Tungkrui, Bangkok 10140, Thailand

ARTICLE INFO

Article history:

Received 30 October 2018

Received in revised form

20 February 2019

Accepted 21 February 2019

Available online 26 February 2019

Keywords:

Shrimp shells

Pyrolysis

Secondary reactions

Cooking treatment

Interactions

ABSTRACT

Shrimp shells discharged in large quantities causes serious environmental pollution. Pyrolysis is a promising method to achieve the resource utilization of discarded shrimp shells as renewable energy. However, the distribution and the characteristics of pyrolysis products are strongly determined by the volatile-char interactions. On the other hand, cooking treatment changes the properties of shrimp shells, making the pyrolysis process more complicated. The focus of this study is to explore the pyrolysis mechanism of both raw and cooked shrimp shells by observing the interactions among volatiles as well as the reactions between volatiles and char. The results showed that char accounted for 60–70 wt% in the pyrolysis products, tar occupied about 20 wt%, and the rest was gases. In the volatile-char interactions, H radicals combined with the special sites to promote the volatilization of the N-containing components and aromatic compounds. Interestingly, a large amount of carbon was stabilized in CaCO₃, which was tightly doped with organic carbon in char. Meanwhile, the secondary reactions among volatiles caused the cyclization and dehydrogenation of the N-containing compounds. After cooking treatment, the partial structures of organics became more easily cracked, and more H radicals went deep into the carbon matrix to generate more volatiles.

© 2019 Elsevier Ltd. All rights reserved.

1. Introduction

Crayfish are crustaceans with exoskeletons containing approximately 30–40 wt% protein, 30–50 wt% mineral salts, 20–30 wt% chitin and other compounds such as pigments and lipids [1]. Huge quantities of crayfish are cultured and cooked especially in central China such as Hubei, Hunan and Anhui provinces. In 2017, the total production of crayfish in China was 11.30 million tons (the year-on-year growth rate was 25.65%), and the shrimp shells waste (including raw and cooked shrimp shells) accounts for 50–70 wt% of the original weight [2,3]. The disorderly disposal of shrimp shells will cause foul odor, water and soil pollution. Therefore, many researchers tried to exploit the shrimp shells to extract chitin and convert it into high-value organic components such as chitosan by using chemical or biological methods [4–6]. However, the

inorganic composition in the shrimp shells was mainly discharged, and a large amount of wastewater was generated by using these methods.

In recent years, it was proposed to realize the resource utilization of discarded shrimp shells through the pyrolysis technology. At medium-low temperatures, the shrimp shells were fully utilized through the decomposition of macromolecular components into carbon materials, tar and pyrolysis gases. In particular, carbon materials from shrimp shells were widely used to generate high-valued products. Zhang et al. [7] used shrimp-shell derived N-doped carbon nanodots as precursors to develop Co/CoO nanoparticles immobilized on Co-N-doped carbon. Based on the self-template of CaCO₃ in the pyrolytic carbon of shrimp shells, Gao et al. [8] prepared nitrogen-decorated hierarchical porous carbon with electrochemical performance. During the pyrolysis of the shrimp cuticle, plenty of volatiles were produced [9]. These volatiles contained useful derivatives such as acetamide, pyridine and pyrrole, which could be applied to the organic synthesis and

* Corresponding author.

E-mail address: hongyunhu@hust.edu.cn (H. Hu).

pharmaceutical industries. Simultaneously, reports on the single-component pyrolysis were particularly common to study high-value utilization methods. Liu et al. [10] evaluated the potential of chitin and chitosan to produce value-added organic nitrogen chemicals like acetamido chemicals and aza-heterocyclic chemicals. Nevertheless, it was dissimilar in the nature of the pyrolysis products prepared by different researchers under various pyrolysis conditions. Through the basic theoretical research on the pyrolysis of shrimp shells, the mechanisms of the influence of pyrolysis conditions on the products could be comprehensively analyzed. Therefore, it will be beneficial to obtain the key technical parameters of the transformation of the shrimp shell pyrolysis products, so as to facilitate the energy utilization of the tar in the later stage, and realize the control of burning pollutants in thermal power generation process by using the char of shrimp shells.

In the pyrolysis process of massive shrimp shells, both of the organic components (represented by chitin and protein) and the inorganic components (mainly composed of calcium carbonate) had a small thermal conductivity, leading to poor heat transfer characteristics especially during the pyrolysis at medium-low temperatures. As a result, the release of volatiles and the char formation might be strongly affected. Further, the secondary reactions among volatiles, as well as the interactions between volatiles and char had a profound effect on the pyrolysis products distribution [11,12]. Gui et al. [13] found that the secondary reaction of volatiles/tar formed during PVC pyrolysis significantly affected the tar formation, for example, the small compounds (such as benzene group and alkanes) from different reactors showed different abundances. Zhang et al.'s research confirmed that the volatile-char interactions could inhibit carbon conversion through three mechanisms: the reaction sites on the carbon surface were occupied by the H radicals, the changes in carbon structure and the volatilization of intrinsic catalysts [14]. Our previous research demonstrated that the H_2S was produced by the hydrogenation-decomposition reactions of sulfonic acid/sulfone/heterocyclic-S, as well as the reactions of inorganic sulfide with H-residue in the char samples [15]. On the other hand, considerable shrimp shells were discharged in the catering industry and cooking treatment was roughly equivalent to a hypoxic pyrolysis process, which might change the structure of the shrimp shells and affect the pyrolysis characteristics of cooked shrimp shells.

In view of the above, this study focused on the pyrolysis of raw shrimp shells and cooked shrimp shells. The experiments were conducted at medium-low temperatures ranging from 400 °C to 600 °C. The interactions among various volatiles or between volatiles and char were investigated by observing the products distribution characteristics at various heating rate from 15 °C·min⁻¹ (by using TGA) to 20000 °C·s⁻¹ (by using Py-GC/MS). More information was obtained through analyzing the thermal decomposition characteristics of derived CaCO_3 and organic carbon materials in the char samples. Moreover, pyrolysis gases and tar were respectively identified to explore the secondary reactions during pyrolysis.

2. Experimental

2.1. Materials

Raw shrimp shells (labeled as RSS) and cooked shrimp shells (labeled as CSS, the cooking process was specifically fried first and then boiled, and the oil temperature was about 200–240 °C) were collected from a local restaurant. Then, the samples were washed, dried and crushed to the particle size of 0.20–0.30 mm. The corresponding experimental and analytical tests were carried out more than 3 times to obtain repeatable results. Table 1 gives the results of proximate and ultimate analyses. The shrimp shells

mainly contained volatiles (more than 60 wt%) and ash (more than 30 wt%). After the cooking treatment, the volatile yield of the shrimp shells increased by 4.78 wt%, and the yields of C, H and N increased.

2.2. Experimental methods

Shrimp shells composition was analyzed using a TGA by heating the samples from 20 °C to 600 °C at 15 °C·min⁻¹ under an inert atmosphere (N_2). Both fast and slow pyrolysis of shrimp shells was carried out in a horizontal tube pyrolysis furnace and maintained for 20 min at each set temperature (400 °C, 500 °C or 600 °C). For the fast pyrolysis, a quartz boat containing a certain sample was placed in the water-cooled portion of the horizontal tube, and the carrier gas (N_2) was input into the horizontal tube reactor to drain the air in the reactor. Then the reactor was heated by the pyrolysis furnace, and the quartz boat was pushed quickly to the horizontal tube when the set temperature was stable. For the slow pyrolysis, the quartz boat was pushed into the reactor at the room temperature and heated to the set temperature at a heating rate of 15 °C·min⁻¹. Finally, after the pyrolysis, the quartz boat was pulled out and the samples were cooled to the room temperature under N_2 . Specifically, volatile products were collected from two pyrolysis experiments under the same conditions. The pyrolytic gases were collected in a gas bag after passing through U-shaped tubes bathed in ice water. Another run was carried out for the collection of tar, simply replacing the iced water with liquid nitrogen [16] (expressed as X-Y, X represents pyrolysis temperature, Y represents fast or slow pyrolysis).

2.3. Analytical methods

Proximate and ultimate analyses of shrimp shells and char samples were performed using a TGA2000 industrial analyzer and a Vario Microcube elemental analyzer, respectively. To obtain the characteristics information of the inorganics and organics in char, the mineralogical structures were analyzed by Empyrean X-ray diffraction (XRD), and the chemical bonds and functional groups of shrimp shell samples as well as the char samples were analyzed using Vertex 70 Fourier transform infrared spectroscopy (FTIR). Moreover, the TGA8000 thermogravimetric analyzer (TGA) was used to characterize the thermal decomposition characteristics of organics and inorganics in char at 30 °C·min⁻¹ under different atmosphere (pure N_2 , N_2 with 20% O_2). To characterize the calorific value of char samples, using a ZDHW-HN 7000B calorimeter to measure their higher heating value (HHV). The pyrolysis gases were identified by GC (Agilent 7890 B), in which N_2 acted as a balancing gas, and the yield of each gas in the pyrolysis products was calculated based on the pyrolysis gas yield and the relative volume fraction of each gas therein. GC/MS (Agilent 7890 A/5975 C) was used to analyze tar compositions.

In order to reduce the influence of secondary reaction on pyrolysis process and pyrolysis products, the fast pyrolysis experiments of shrimp shells were performed by using an analytical Py-GC/MS instrument, in which GC/MS (Clarus 560) was coupled to a pyrolyzer (CDS 5200HP). The prepared shrimp shell samples were weighed using an analytical balance with a readability of 0.01 mg, and the quantity in each sample was precisely controlled to 0.30 mg. The sample of the weighed shrimp shell was placed in the middle of the quartz tube. In order to prevent the escape of solid particles during the pyrolysis process, quartz wool was placed on both sides of the shrimp shell sample. The quartz tube containing the shrimp shell sample was placed in a pyrolyzer, and the fast pyrolysis was achieved by utilizing the rapid heating characteristics of the platinum wire. The fast pyrolysis experiments were carried

Table 1
Properties of raw and cooked shrimp shells.

Samples	Proximate analysis (%) ^a			Ultimate analysis (%)				
	Volatile matter	Ash	Fixed carbon	C	H	O ^b	N	S
RSS	62.66	37.29	0.05	22.60	2.91	33.47	3.49	0.24
CSS	67.44	32.43	0.13	27.82	4.01	30.87	4.61	0.26

^a Dry basis.

^b Calculated by difference.

out from 50 °C to different temperatures (400 °C, 600 °C), and the heating rates were set to 200 °C·s⁻¹ and 20000 °C·s⁻¹, respectively. Chromatographic separation was achieved by using an Elite-35 MS capillary column (30 m × 0.25 mm i.d., 0.25 μm film thickness), and the pyrolysis volatiles were analyzed in real time by GC/MS. Helium (99.999%) was a carrier gas with a constant flow rate of 1 mL min⁻¹ at a split ratio of 1:80. The GC furnace was heated from 40 °C to 160 °C at a heating rate of 10 °C·min⁻¹, then heated to 280 °C at a heating rate of 15 °C·min⁻¹. For each sample, at least three independent experiments were performed to ensure repeatable data results. In addition, various chromatographic peaks were identified based on the NIST library. The chromatographic peak area of the compound was linearly related to its content, and the relationship between the relative peak area and the concentration of the detected pyrolysis product was also linear. Therefore, the peak area values (%) were compared to evaluate the yield of each component in the tar under different conditions. More detailed information was mentioned elsewhere [17].

3. Results and discussion

3.1. Yield distribution of pyrolysis products

Fig. 1 shows the TG and DTG curves of shrimp shells. Obviously, the weight of RSS was reduced by 38.84 ± 1.89 wt%, and CSS lost more weight (44.04 ± 2.28 wt%). Due to the volatilization of water and the decomposition of some lipids, the temperature range of the first weight loss was concentrated before 200 °C. The second weight loss ranging from 262 °C to 421 °C was related to the decomposition of protein and chitin. Notably, the DTGmax (the absolute value of the maximum weight loss rate) of RSS was lower than CSS. It might be that the cooking treatment broke some chemical bonds of organics in shrimp shells and caused protein denaturation, resulting in some easily decomposed structures in CSS.

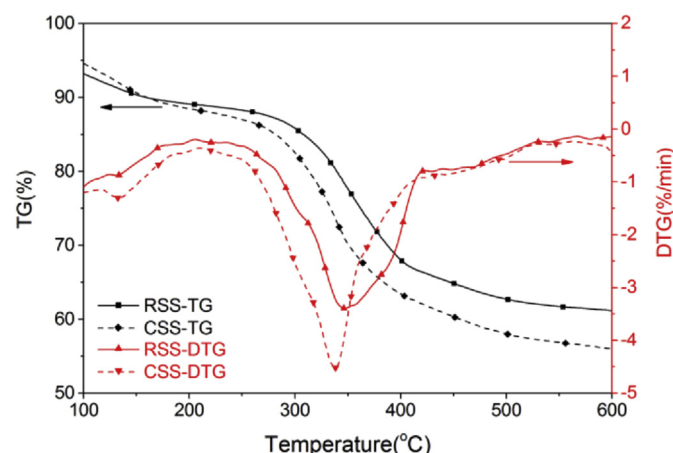


Fig. 1. TGA and DTG curves of shrimp shells samples.

The distribution of pyrolysis products is displayed in Fig. 2. It was intuitively seen that char accounted for 60–70 wt%, tar occupied about 20 wt% and pyrolysis gases accounted for 10–15 wt%. When the temperature rose from 400 °C to 600 °C, the char yield reduced by about 7 wt% and the gas yield increased by about 7 wt% in the pyrolysis of RSS (Fig. 2(a)). High temperature contributed to the breaking of chemical bonds and the cleavage of macromolecular compounds [18,19], and the decrease in the char yield was also ascribed to the secondary decomposition of the carbon residue. Simultaneously, the heating rate played an important role in the pyrolysis process. Fast pyrolysis reduced heat and mass transfer limitations and shortened the available time for secondary

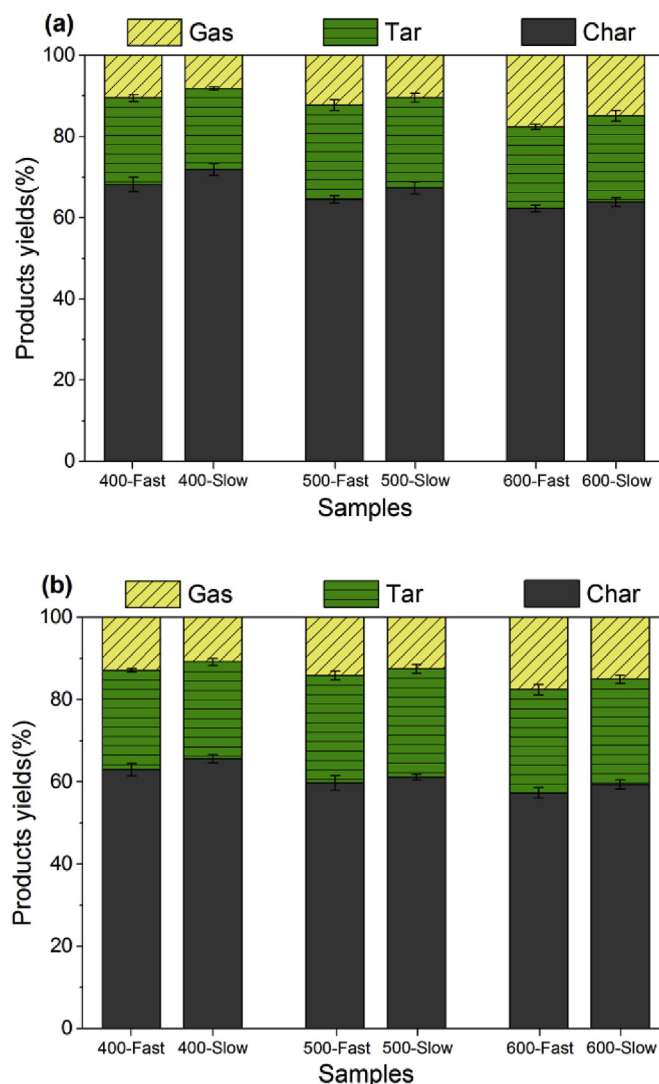


Fig. 2. Products yields during the pyrolysis of (a) RSS and (b) CSS.

reactions, which facilitated organics cracking to produce more volatiles and less char [20,21]. Interestingly, the yields of char and pyrolysis gases were negative correlativity under the influence of different temperatures and heating rates.

As shown in Fig. 2(b), the similar product distribution was also found in CSS. Dissimilarity, the cooking treatment could be equivalent to an advanced hypoxic pyrolysis process, which reduced the stability of the organic components in the shrimp shells and in turn affected the product distribution (char yield was reduced by approximately 5 wt%). Furthermore, the organic condiments were involved in the pyrolysis reactions, and the organic components in the shrimp shells might react twice on the surface of inorganic condiments to produce more light compounds [22,23].

3.2. Formation mechanism of char from shrimp shells

As Fig. 3 depicts, the mineralogical characteristics of char were determined by XRD, the peak shapes of different char samples were very analogous. CaCO_3 was the only crystal detected, and its phases were mainly calcite with a small fraction of aragonite. The information about inorganics and organics in different char samples are summarized in Table 2. Combined the ash contents and the CaCO_3 contents in the char samples, it could be confirmed that CaCO_3 was

the main fixed component in char. The high content of CaCO_3 also confirmed that the HHV of char was low, so the potential for combustion of shrimp shell char as a direct fuel was small. However, they still had recyclable value, such as controlling the pollution gases and trace elements in the combustion process. Activated CaCO_3 could achieve physical and chemical adsorption of heavy metals. Previous researches showed that CaO could effectively capture arsenic and then produced relatively stable compounds at high temperatures [24,25]. At the same time, part of the SO_2 in the flue gas was removed [24]. In the process of controlling pollutants, the calorific value of residual organics (the part other than ash) in char were available. It was the volatilization of organics that affected the proportion of ash and CaCO_3 in char. The residual organics were reduced by roughly 10 wt% when the temperature increased from 400 °C to 600 °C, and the effect of increasing the heating rate was not so obvious (reduced by about 4 wt%). Moreover, the fast pyrolysis at high temperature promoted the volatilization of N-containing components in organics.

In order to explore the relevant changes in chemical bonds and functional groups in the samples under different pyrolysis conditions, Fig. 4 shows the FTIR spectra of shrimp shell and char samples. The absorption peaks of CO_3^{2-} at 1413 cm^{-1} , 874 cm^{-1} and 714 cm^{-1} respectively represented the anti-symmetric stretching vibration, the out-of-plane bending vibration and the in-plane bending vibration, they were all characteristic peaks of calcite. Organic calcium gradually transformed into CaCO_3 during pyrolysis, thus the absorption peaks of CO_3^{2-} in the char samples were more obvious. The overlapping peak between 1000 and 1200 cm^{-1} referred to C–O–C functional groups [26–28], these functional groups came from the furan rings accumulated in char after pyrolysis, and the presence of furan derivatives could also be detected by GC-MS analysis. Moreover, the stretching vibration region of OH was observed at 3300–3500 cm^{-1} in all samples [29–31], which indicated that a part of the structure of alcohols and phenols were present in the char samples after the pyrolysis process. The stretching and bending of CH at 2800 cm^{-1} and 3000 cm^{-1} showed the presence of alkanes, as the pyrolysis temperature rose to 600 °C, this part of the organics gradually changed and volatilized. In addition, some N-containing functional groups in the chitin and protein converted into volatiles after pyrolysis, thus the absorption peak of NH (the bending vibration at 1659 cm^{-1} [32]) was only detected in the shrimp shell samples. The above discussion indicated that the organics underwent complex decomposition and reforming during the pyrolysis process, and some residual organics were aggregated in the char with the inorganics represented by CaCO_3 . However, the differences of FTIR spectra between RSS and CSS and among different char samples were not obvious, the further exploration was carried out through the thermogravimetric analysis.

The DTG curves for char samples performed in an inert atmosphere are shown in Fig. 5. According to Fig. 5(a), the char prepared at 400 °C was divided into two weight loss peaks when the temperature rose from 200 °C to 900 °C. The first weight loss peak around 450 °C contained the cracking of partial residual N-containing organics. The second weight loss peak, occurred between 611 °C and 769 °C, was mainly caused by the decomposition of CaCO_3 to release CO_2 (Eq. (1)). Comparing the decomposition of the calcite in char with the decomposition of pure calcite, their DTG curves were similar, but the decomposition temperature of the calcite in char was reduced by 100–250 °C [33]. Related studies reported that biotic calcite was more susceptible to decomposition than abiotic calcite, which was attributed to the fact that biotic calcite was structurally more fragile [34]. On the other hand, well-crystallised calcite had a higher decomposition temperature than amorphous and metastable calcium carbonate polymorphs [35]. In

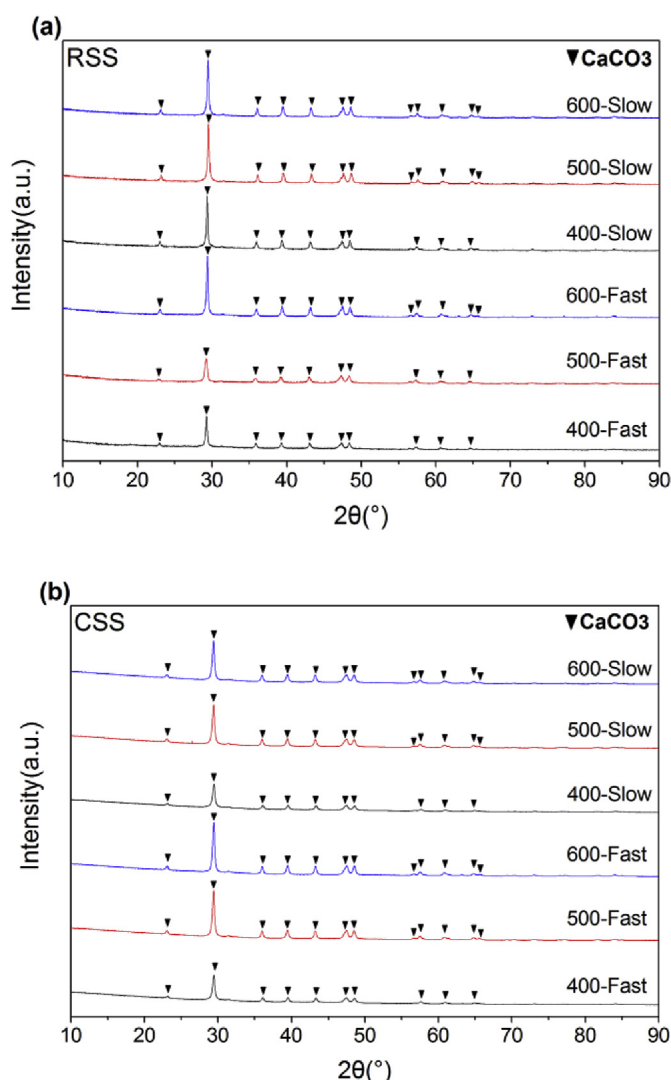


Fig. 3. XRD patterns of the char samples from (a) RSS and (b) CSS.

Table 2
Properties of char samples derived from different pyrolysis conditions.

Samples	Ash in char (wt%) ^a	CaCO ₃ in char (wt%) ^b	Organics in char (wt%) ^c	N in organics (wt%) ^d	HHV (MJ/kg)
RSS-400-Fast	63.53	55.50	36.47	5.14	2.73
RSS-400-Slow	57.20	50.53	42.80	5.62	2.93
RSS-600-Fast	73.54	68.86	26.46	4.81	1.84
RSS-600-Slow	70.25	65.34	29.75	5.37	2.05
CSS-400-Fast	64.23	54.45	35.77	5.95	2.44
CSS-400-Slow	60.14	52.82	39.86	6.43	2.69
CSS-600-Fast	75.52	70.95	24.48	5.26	1.77
CSS-600-Slow	71.08	67.36	28.92	5.68	1.89

^a Burning char to ash at 550 °C in the muffle furnace.

^b Estimated content based on CO₂ produced by decomposition of CaCO₃ at high temperature in TG measurements.

^c Calculated by difference.

^d Determined by ultimate analysis.

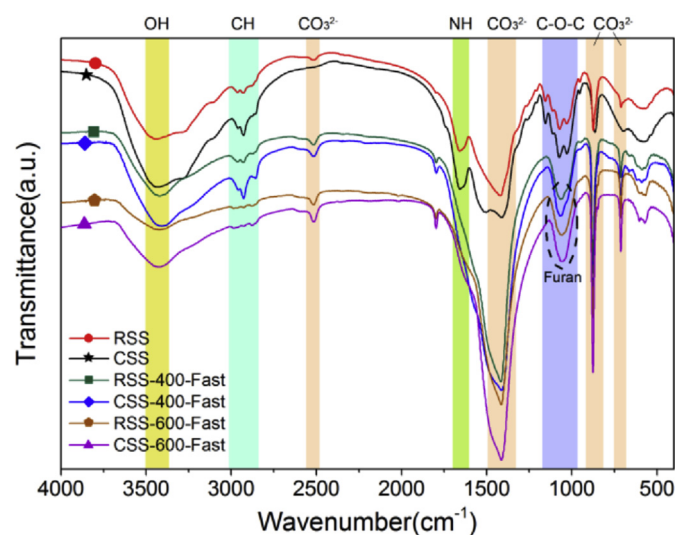


Fig. 4. FTIR spectra of shrimp shells samples and char samples.

the process of pyrolysis of shrimp shells, the precursor of CaCO₃ were doped with organics, these impurities affected the crystallization of CaCO₃, leading to the generation of crystal defects. In addition, as can be seen from Fig. 5(b) and Table 2, the char prepared at 600 °C had more CaCO₃, and there was a similar trend after the cooking treatment, thus the DTGmax of CaCO₃ was higher. After the cooking treatment, the decomposition temperature of CaCO₃ in char was reduced by about 20 °C, which might be caused by the smaller particle size of CaCO₃. It was speculated that the adsorption performance of CSS's char during combustion was better. Because the CO₂ was released in advance, and the adsorbates were prevented to form a dense layer on the surface of CaO.

In order to analyze the decomposition characteristics of char under air atmosphere, Fig. 5 shows the DTG curves of char in an oxidizing atmosphere (in N₂ with 20% O₂). Apart from the cracking of the easily decomposable organic components (Peak 2 in Fig. 5(c) and (d)) and CaCO₃ (Peak 3), the oxidation of the organics in the char samples was observed (Peak 1). Meanwhile, the yield of carbon-hydrogen bonds in the char samples prepared at 400 °C was more abundant (Fig. 4), resulting in organics were more susceptible to be oxidized (Peak 1 appeared early). Another major change was the decomposition temperature of CaCO₃ became higher in the oxidizing atmosphere, which was caused by the thermodynamic equilibrium reaction (Eq. (1); Eq. (2)), the change in oxygen partial pressure increased the decomposition temperature of CaCO₃.



3.3. Formation mechanism of volatiles from shrimp shells

Accompanied by the char formation process, small molecule gases and macromolecular tar were simultaneously produced. As seen in Fig. 6, the pyrolysis gases of shrimp shells were mainly CO₂, H₂ and CH₄. During the pyrolysis process, the effect of temperature on the three gas contents was particularly pronounced. CO₂ was chiefly derived from the further cleavage of the C–O–C bonds after the glycosidic bonds in chitin were destroyed [36]. When chitin was pyrolyzed to produce 2-methylpyridine, 3-methylpyrrole, acetamide and 2-methylfuran, there were accompanied by the release of CO₂ (Eq. (3); Eq. (4); Eq. (5)). As the temperature increased, these reactions were partially inhibited, or CO₂ might interact with semi-coke to cause a decrease in yield. The generation of H₂ mainly came from three aspects, chitin was pyrolyzed to produce N-heterocycles (Eq. (3)), the interactions of acetamides with other volatiles (Eq. (6)), and the removal of R groups from specific amino acids (Eq. (9)). The yield of H₂ increased significantly due to the intensification of these reactions under fast pyrolysis at raised temperature up to 600 °C. Some CH₄ was released during the chitin pyrolysis process (Eq. (4)), and the secondary decomposition of volatiles also produced CH₄ as the temperature increased to 500–600 °C [37]. In addition, high heating rate increased the production of pyrolysis gases and promoted the secondary reactions among volatiles, resulting in increasing yields for each gas.

The secondary reactions in pyrolysis were effectively inhibited due to the short residence time by Py-GC/MS [38]. In order to analyze primary tar, the typical components of tar from fast pyrolysis are given in Table 3 and Table 4. The main components in primary tar were acetamide, toluene, 3-acetamido-5-methylfuran, phenol, p-cresol and oxazoline derivatives. A lot of derivatives derived from chitin pyrolysis were found, such as 4-methyl-pyridine (Eq. (3)), 3-methylpyrrole (Eq. (4)), acetamide (Eq. (5)), 3-acetamido-5-methylfuran (Eq. (6)) and oxazoline derivatives. Meanwhile, derivatives produced by protein pyrolysis also exhibited great abundance, and they could be assigned to specific amino acid sequences or specific amino acid sequences of polypeptides [39], such as toluene were from phenylalanine (Eq. (7)), phenol was from tyrosine (Eq. (8)), pyrrole was from proline (Eq. (9)) and indole was from tryptophan (Eq. (10)). Moreover, there were some pyrolysis derivatives derived from non-chitin polysaccharide, such as 3-methyl-2-cyclopenten-1-one and 2-methyl-2-cyclopenten-1-one. These results agreed with the relevant references [40,41]. On

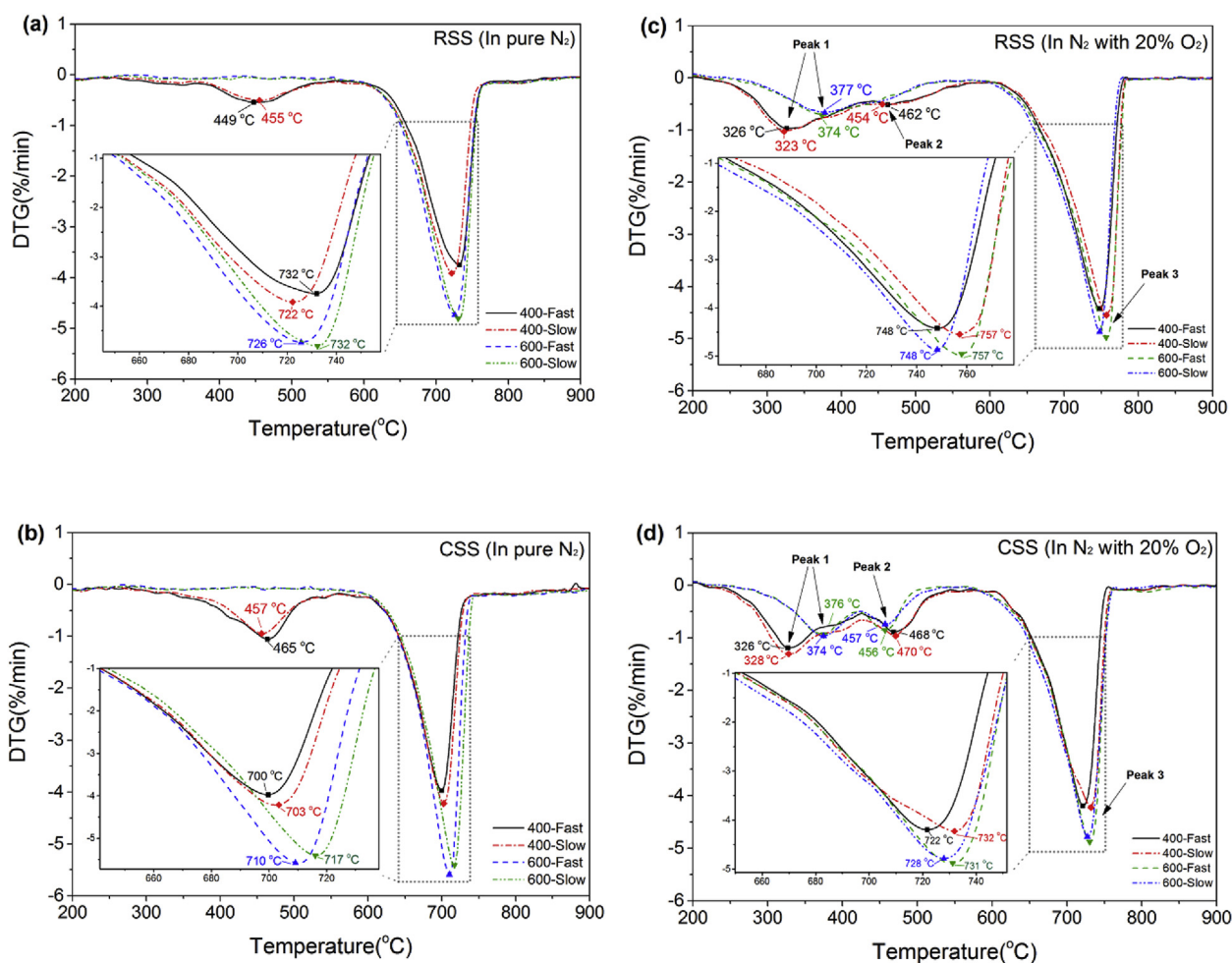
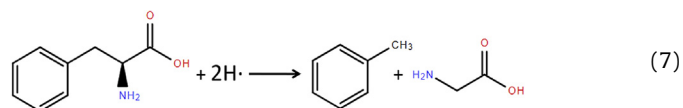
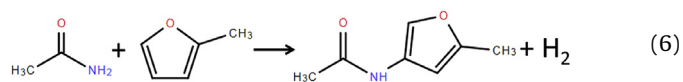
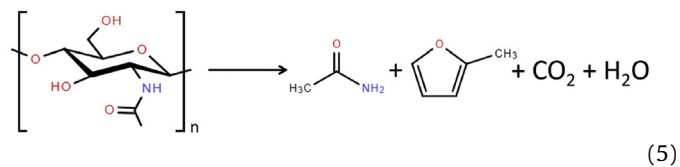
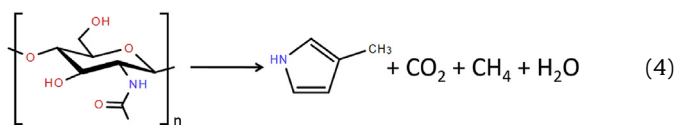
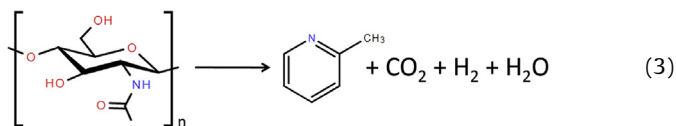


Fig. 5. DTG curves for the decomposition of char samples in the carrier gas with different oxygen concentrations.

the other hand, according to a large amount of formulas for calculating the calorific value by using fuel element analysis [42], it was estimated that the HHV of the shrimp shell tar was between 11 and 15 MJ/kg, and the HHV of the CSS's tar was higher. After further chemical modification and activation treatment, if the transfer of N can be effectively controlled, the tar has the potential to be biofuels. Simultaneously, the main component of tar, acetamide, can be isolated and extracted for use in the chemical, plastic, pharmaceutical and other industries.



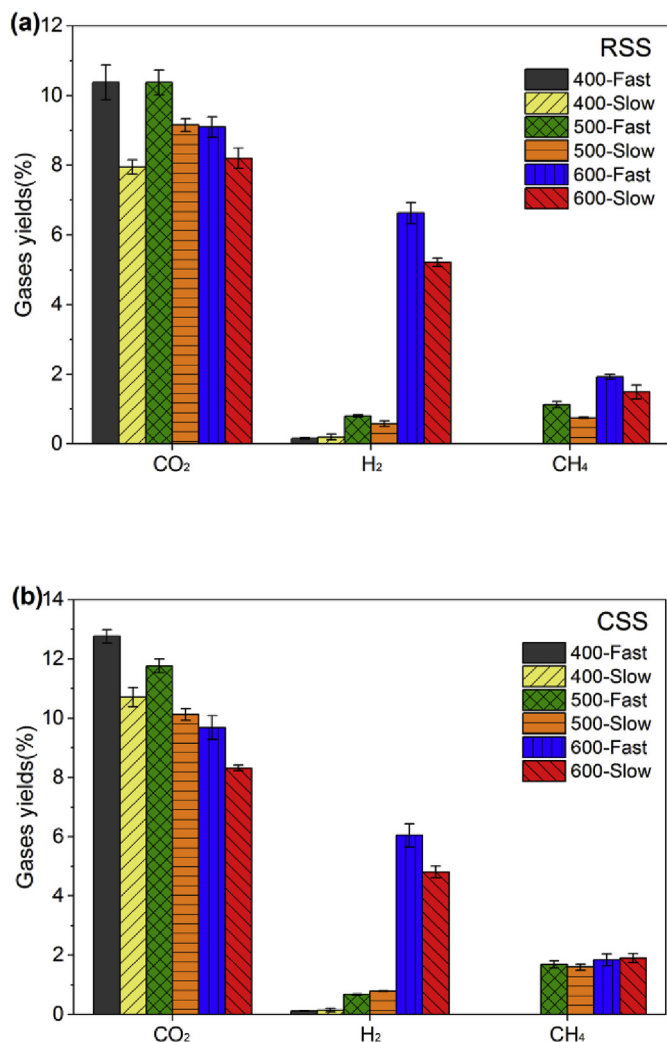


Fig. 6. The distribution of gases during the pyrolysis of (a) RSS and (b) CSS.

Table 3

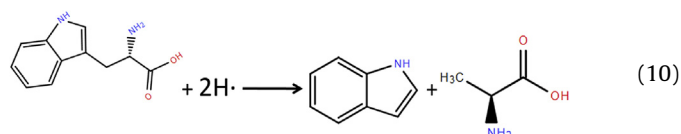
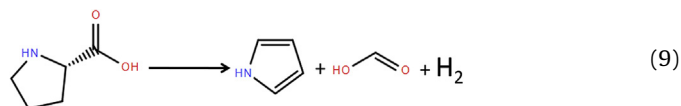
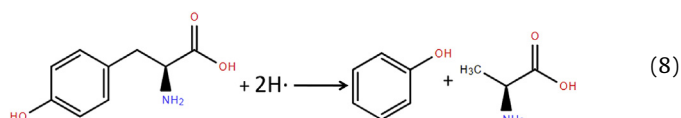
Typical tar components from fast pyrolysis of RSS by Py-GC/MS.

Compound name	Formula	Area (%)			
		200 °C · s ⁻¹		20000 °C · s ⁻¹	
		400 °C	600 °C	400 °C	600 °C
Acetic acid	C ₂ H ₄ O ₂	0.17	0.30	0.24	0.33
Acetohydroxamic acid	C ₂ H ₅ NO ₂	1.47	1.75	1.69	2.08
Toluene	C ₇ H ₈	3.03	9.24	4.23	9.53
Pyrrole	C ₄ H ₅ N	1.70	4.12	1.97	4.82
Pyridine	C ₅ H ₅ N	0.28	0.83	0.68	1.02
4-Methyl-pyridine	C ₆ H ₇ N	1.09	1.51	0.56	1.26
Acetamide	C ₂ H ₅ NO	27.69	28.21	31.43	31.84
3-Methylpyrrole	C ₅ H ₇ N	1.01	2.93	1.63	2.99
Phenylethylene	C ₈ H ₈	1.08	2.06	0.96	2.01
Pyridone derivative	- ^a	1.06	1.22	1.11	0.84
2-Methyl-2-cyclopenten-1-one	C ₆ H ₈ O	1.63	2.39	1.63	1.85
Phenol	C ₆ H ₆ O	4.35	5.23	4.07	5.01
3-Methyl-2-cyclopenten-1-one	C ₆ H ₈ O	0.43	1.27	1.66	1.22
3-Methyl-butanamide	C ₅ H ₁₁ NO	0.73	0.76	0.88	0.76
Ethylpyrrole	C ₆ H ₉ N	4.19	2.82	3.76	2.52
p-Cresol	C ₇ H ₈ O	4.61	6.28	3.84	6.15
3-Amino-5-methylpyrazole	C ₄ H ₇ N ₃	0.40	1.11	0.91	1.05
2,4-Dimethylphenol	C ₈ H ₁₀ O	2.48	2.18	2.45	2.45
Ethylcyanobenzene	C ₉ H ₉ N	1.04	1.59	1.36	1.42
N-methyl-N-ethenylacetamide	C ₅ H ₉ NO	2.98	2.63	2.33	2.45
3-Acetamidofuran	C ₆ H ₇ NO ₂	2.13	1.28	1.73	1.16
Benzenepropanenitrile	C ₉ H ₉ N	1.06	1.23	0.86	1.14
1,1'-Methylenebis-piperidine	C ₁₁ H ₂₂ N ₂	4.11	2.40	3.29	2.06
Indole	C ₈ H ₇ N	3.09	1.98	2.73	1.73
3-Acetamido-5-methylfuran	C ₇ H ₉ NO ₂	12.23	4.83	10.21	3.72
Oxazoline derivatives	- ^a	7.61	4.05	6.17	3.64
Palmitic acid	C ₁₆ H ₃₂ O ₂	2.97	1.94	2.89	1.55
2,5-Diketopiperazine	C ₄ H ₆ N ₂ O ₂	1.91	0.75	1.22	0.70
5,9-Dimethyl-2-decanone	C ₁₂ H ₂₄ O	1.31	0.57	1.01	0.48
Oleic acid	C ₁₈ H ₃₄ O ₂	0.30	0.93	1.30	0.82
Lauramide	C ₁₂ H ₂₅ NO	1.29	0.79	0.96	0.67
Oleamide	C ₁₈ H ₃₅ NO	0.56	0.83	0.22	0.72

^a Unable to represent.

components in a part of the amino acids. The N-heterocycles (such as pyrrole and pyridine derivatives) in the primary tar were mainly derived from the reforming of acetyl glucosamine structure in chitin and the detachment of the R groups of the amino acid. Meanwhile, the volatile-char interactions were accompanied by the destruction of the organic structure, and the essence of the volatile-char interactions was the reactions between H radicals and char [11]. Some H radicals occupied special binding sites and promoted the decomposition of amino acids (Eq. (7); Eq. (8); Eq. (10)). As the temperature rose from 400 °C to 600 °C, the yield of aromatic compounds increased, which was related that H radicals penetrated into char to promote aromatization, and increasing the heating rate had similar effects. The char matrix became “loose” after the cooking treatment, thus H radicals were more likely to penetrate deeper into char to produce more volatiles.

The N-containing compounds in primary tar didn't exhibit significant conversion characteristics due to the decrease of the secondary reactions. For further analysis, the tar collected after pyrolysis was detected by GC/MS, Fig. 7 shows the distribution of N-containing compounds and aromatic compounds in tar. In the pyrolysis process, the decomposition and reforming of tar increased the release of H radicals. Simultaneously, H radicals could affect the conversion and selectivity of N-containing components [43], which combined with the activated N sites to promote the detachment of the N-containing components. Combined with Fig. 7 and Table 2, the N-containing components had a



According to the results in Tables 3 and 4, it could be clearly found that acetamide was the most abundant in the primary tar (about 30% in tar), which mainly came from the breakage of the C–N bonds in chitin and the detachment of the N-containing

Table 4

Typical tar components from fast pyrolysis of CSS by Py-GC/MS.

Compound name	Formula	Area (%)			
		200 °C · s ⁻¹		20000 °C · s ⁻¹	
		400 °C	600 °C	400 °C	600 °C
Acetic acid	C ₂ H ₄ O ₂	0.22	0.25	0.20	0.25
Acetohydroxamic acid	C ₂ H ₅ NO ₂	4.68	3.57	5.50	3.72
Toluene	C ₇ H ₈	8.41	11.16	8.49	12.24
Pyrrole	C ₄ H ₅ N	2.80	4.92	3.12	5.11
Pyridine	C ₅ H ₅ N	0.83	0.67	1.21	1.17
4-Methyl-pyridine	C ₆ H ₇ N	0.85	1.13	0.91	1.55
Acetamide	C ₂ H ₅ NO	32.13	29.57	32.57	27.72
3-Methylpyrrole	C ₅ H ₇ N	1.37	2.26	1.46	2.39
Phenylethylene	C ₈ H ₈	1.55	1.59	0.88	1.98
Pyridone derivative	- ^a	0.90	0.41	0.99	0.72
2-Methyl-2-cyclopenten-1-one	C ₆ H ₈ O	1.87	1.40	1.82	1.91
Phenol	C ₆ H ₆ O	4.14	5.46	4.65	5.40
3-Methyl-2-cyclopenten-1-one	C ₆ H ₈ O	1.33	0.97	1.21	0.83
3-Methyl-butanamide	C ₅ H ₁₁ NO	1.19	1.13	1.23	0.90
Ethylpyrrole	C ₆ H ₉ N	2.51	2.77	2.39	2.63
p-Cresol	C ₇ H ₈ O	3.99	5.69	3.87	5.74
3-Amino-5-methylpyrazole	C ₄ H ₇ N ₃	0.68	0.84	0.68	0.79
2,4-Dimethylphenol	C ₈ H ₁₀ O	2.05	1.54	1.30	1.97
Ethylcyanobenzene	C ₉ H ₉ N	0.90	1.47	0.77	1.52
N-methyl-N-ethenylacetamide	C ₅ H ₉ NO	2.04	1.52	2.04	1.57
3-Acetamidofuran	C ₆ H ₇ NO ₂	1.11	0.87	1.13	0.81
Benzenepropanenitrile	C ₉ H ₉ N	1.57	1.63	1.63	1.55
1,1'-Methylenebis-piperidine	C ₁₁ H ₂₂ N ₂	2.75	2.10	2.68	1.93
Indole	C ₈ H ₇ N	3.15	3.23	3.08	3.00
3-Acetamido-5-methylfuran	C ₇ H ₉ NO ₂	5.14	3.27	4.96	2.68
Oxazoline derivatives	- ^a	3.73	3.25	3.73	2.82
Palmitic acid	C ₁₆ H ₃₂ O ₂	2.50	1.81	2.56	1.86
2,5-Diketopiperazine	C ₄ H ₆ N ₂ O ₂	0.66	0.44	0.60	0.50
5,9-Dimethyl-2-decanone	C ₁₂ H ₂₄ O	2.06	0.82	2.10	0.79
Oleic acid	C ₁₈ H ₃₄ O ₂	1.04	1.84	0.87	1.68
Lauramide	C ₁₂ H ₂₅ NO	0.77	0.89	0.77	0.85
Oleamide	C ₁₈ H ₃₅ NO	1.07	1.51	0.58	1.42

^a Unable to represent.

tendency to crack and volatilize from char during the fast pyrolysis at high temperature. Moreover, the secondary reactions among the volatiles caused the cyclization and dehydrogenation of the N-

containing components, thus more N-heterocycles and H₂ were produced. The dehydrogenation phenomenon was more obvious due to the interactions among volatiles under fast pyrolysis. High temperatures promoted the cleavage of the C-R bonds in the amino acids, thus the release of the aromatic compounds increased. On the other hand, H radicals could penetrate into carbon matrix to induce the aromatization and the condensation of aromatic rings [44]. The abundance of these tar components were significantly increased after the cooking treatment, which confirmed the results of the previous discussion.

4. Conclusions

In the present study, slow and fast pyrolysis experiments of both raw and cooked shrimp shells were studied in the temperature range of 400 °C–600 °C. The pyrolysis mechanisms of shrimp shells were discussed based on the interactions among volatiles and the reactions between volatiles and char. In the yield of pyrolysis products, char (~60–70 wt%) > tar (~20 wt%) > gases (~10–15 wt%). As the heating rate increased, the light components (especially N-containing compounds) in char gradually volatilized, and the process was accompanied by the production of N-heterocycles. When the temperature increased from 400 °C to 600 °C, the glycosidic bonds and the C–N bonds in chitin, as well as the C-R bonds in amino acids were more susceptible to cleavage to produce N-heterocycles and aromatics. In the interactions of pyrolysis, H radicals penetrated into the char matrix and combined with the special sites to promote the volatilization of the N-containing components and aromatic compounds, which also altered the structure and properties of carbon. The cooking treatment could be equivalent to an advanced hypoxic pyrolysis process, the stability of the chemical bonds became poor, thus producing more tar and gases under the same pyrolysis conditions. In addition, cooked shrimp shells are more likely to be renewable energy sources, that is, char controls pollutants in the combustion process, and tar is used in biofuels and chemical raw materials.

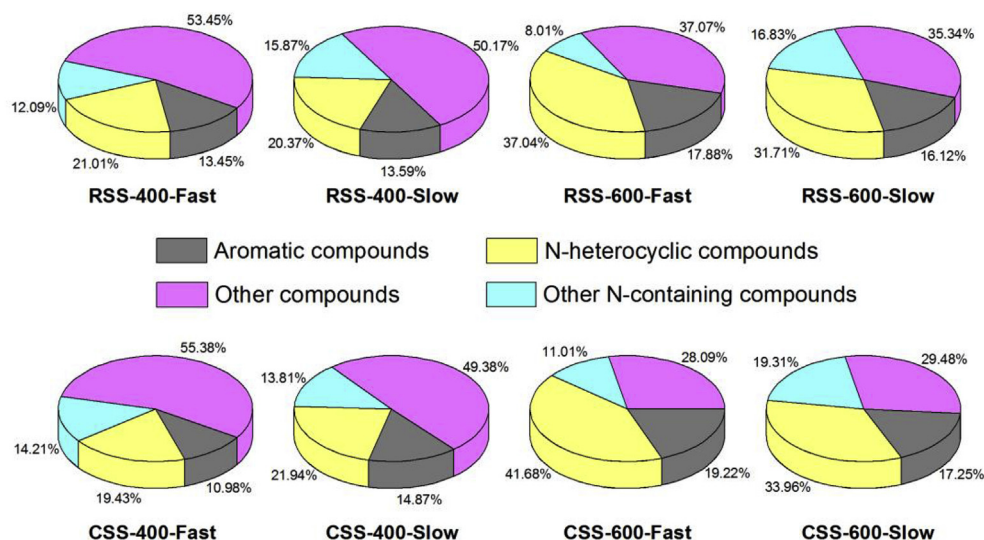


Fig. 7. The tar components from the pyrolysis of shrimp shells by GC/MS.

Acknowledgments

This work was financially supported by the National Natural Science Foundation of China (51606075, 51661145010) and Major Technical Innovation Projects of Hubei Province, China (2018ACA151, 2018CKB906). We are extremely grateful to the Analytical and Testing Center of Huazhong University of Science and Technology for the experimental measurements.

References

- [1] Y.I. Cho, H.K. No, S.P. Meyers, Physicochemical characteristics and functional properties of various commercial chitin and chitosan products, *J. Agric. Food Chem.* 46 (1998) 3839–3843, <https://doi.org/10.1021/jf971047f>.
- [2] Saima M. Kuddus, Roohi I.Z. Ahmad, Isolation of novel chitinolytic bacteria and production optimization of extracellular chitinase, *J. Genet. Eng. Biotechnol.* 11 (2013) 39–46, <https://doi.org/10.1016/j.jgeb.2013.03.001>.
- [3] M.S. Islam, S. Khan, M. Tanaka, Waste loading in shrimp and fish processing effluents: potential source of hazards to the coastal and nearshore environments, *Mar. Pollut. Bull.* 49 (2004) 103–110, <https://doi.org/10.1016/j.marpolbul.2004.01.018>.
- [4] R.S. Juang, F.C. Wu, R.L. Tseng, Solute adsorption and enzyme immobilization on chitosan beads prepared from shrimp shell wastes, *Bioresour. Technol.* 80 (2001) 187–193, [https://doi.org/10.1016/S0960-8524\(01\)00090-6](https://doi.org/10.1016/S0960-8524(01)00090-6).
- [5] K.N. Aye, R. Karuppuswamy, T. Ahamed, W.F. Stevens, Peripheral enzymatic deacetylation of chitin and reprecipitated chitin particles, *Bioresour. Technol.* 97 (2006) 577–582, <https://doi.org/10.1016/j.biortech.2005.03.030>.
- [6] N. Pacheco, M. Garnica-González, J.Y. Ramírez-Hernández, B. Flores-Albino, M. Gimeno, E. Bázana, K. Shirai, Effect of temperature on chitin and astaxanthin recoveries from shrimp waste using lactic acid bacteria, *Bioresour. Technol.* 100 (2009) 2849–2854, <https://doi.org/10.1016/j.biortech.2009.01.019>.
- [7] X. Zhang, R. Liu, Y. Zang, G. Liu, G. Wang, Y. Zhang, H. Zhang, H. Zhao, Co/CoO nanoparticles immobilized on Co-N-doped carbon as trifunctional electrocatalysts for oxygen reduction, oxygen evolution and hydrogen evolution reactions, *Chem. Commun.* 52 (2016) 5946–5949, <https://doi.org/10.1039/c6cc02513g>.
- [8] F. Gao, J. Qu, C. Geng, G. Shao, M. Wu, Self-templating synthesis of nitrogen-decorated hierarchical porous carbon from shrimp shell for supercapacitors, *J. Mater. Chem. A* 4 (2016) 7445–7452, <https://doi.org/10.1039/c6ta01314g>.
- [9] N.S. Gupta, R. Michels, D.E.G. Briggs, R.P. Evershed, R.D. Pancost, The organic preservation of fossil arthropods: an experimental study, *Proc. R. Soc. B Biol. Sci.* 273 (2006) 2777–2783, <https://doi.org/10.1098/rspb.2006.3646>.
- [10] C. Liu, H. Zhang, R. Xiao, S. Wu, Value-added organonitrogen chemicals evolution from the pyrolysis of chitin and chitosan, *Carbohydr. Polym.* 156 (2017) 118–124, <https://doi.org/10.1016/j.carbpol.2016.09.024>.
- [11] C.Z. Li, Importance of volatile-char interactions during the pyrolysis and gasification of low-rank fuels – a review, *Fuel* 112 (2013) 609–623, <https://doi.org/10.1016/j.fuel.2013.01.031>.
- [12] J.D. Martínez, N. Puy, R. Murillo, T. García, M.V. Navarro, A.M. Mastral, Waste tyre pyrolysis – a review, *Renew. Sustain. Energy Rev.* 23 (2013) 179–213, <https://doi.org/10.1016/j.rser.2013.02.038>.
- [13] B. Gui, Y. Qiao, D. Wan, S. Liu, Z. Han, H. Yao, M. Xu, Nascent tar formation during polyvinylchloride (PVC) pyrolysis, *Proc. Combust. Inst.* 34 (2013) 2321–2329, <https://doi.org/10.1016/j.proci.2012.08.013>.
- [14] S. Zhang, J.I. Hayashi, C.Z. Li, Volatilisation and catalytic effects of alkali and alkaline earth metallic species during the pyrolysis and gasification of Victorian brown coal. Part IX. Effects of volatile-char interactions on char-H₂O and char-O₂ reactivities, *Fuel* 90 (2011) 1655–1661, <https://doi.org/10.1016/j.fuel.2010.11.008>.
- [15] H. Liu, L. Yi, Q. Zhang, H. Hu, G. Lu, A. Li, H. Yao, Co-production of clean syngas and ash adsorbent during sewage sludge gasification: synergistic effect of Fenton peroxidation and CaO conditioning, *Appl. Energy* 179 (2016) 1062–1068, <https://doi.org/10.1016/j.apenergy.2016.07.063>.
- [16] H. Liu, Q. Zhang, H. Hu, P. Liu, X. Hu, ScienceDirect Catalytic role of conditioner CaO in nitrogen transformation during sewage sludge pyrolysis, *Proc. Combust. Inst.* 35 (2015) 2759–2766, <https://doi.org/10.1016/j.proci.2014.06.034>.
- [17] Z.B. Zhang, Q. Lu, X.N. Ye, W.T. Li, B. Hu, C.Q. Dong, Production of phenolic-rich bio-oil from catalytic fast pyrolysis of biomass using magnetic solid base catalyst, *Energy Convers. Manag.* 106 (2015) 1309–1317, <https://doi.org/10.1016/j.enconman.2015.10.063>.
- [18] Z. Yu, M. Dai, M. Huang, S. Fang, J. Xu, Y. Lin, X. Ma, Catalytic characteristics of the fast pyrolysis of microalgae over oil shale: analytical Py-GC/MS study, *Renew. Energy* 125 (2018) 465–471, <https://doi.org/10.1016/j.renene.2018.02.136>.
- [19] H. Zhang, J. Hu, J. Zheng, R. Xiao, Fast pyrolysis of cellulosic ethanol solid residues with high ash and lignin contents in a fluidized bed, *J. Anal. Appl. Pyrolysis* 128 (2017) 217–223, <https://doi.org/10.1016/j.jaap.2017.10.008>.
- [20] J. Akhtar, N. Saidina Amin, A review on operating parameters for optimum liquid oil yield in biomass pyrolysis, *Renew. Sustain. Energy Rev.* 16 (2012) 5101–5109, <https://doi.org/10.1016/j.rser.2012.05.033>.
- [21] Z. Huang, Z. Deng, D. Chen, F. He, S. Liu, K. Zhao, G. Wei, A. Zheng, Z. Zhao, H. Li, Thermodynamic analysis and kinetic investigations on biomass char chemical looping gasification using Fe-Ni bimetallic oxygen carrier, *Energy* 141 (2017) 1836–1844, <https://doi.org/10.1016/j.energy.2017.11.127>.
- [22] D. Carpenter, T.L. Westover, S. Czernik, W. Jablonski, Biomass feedstocks for renewable fuel production: a review of the impacts of feedstock and pre-treatment on the yield and product distribution of fast pyrolysis bio-oils and vapors, *Green Chem.* 16 (2014) 384–406, <https://doi.org/10.1039/c3gc41631c>.
- [23] K.P. Shadangi, K. Mohanty, Thermal and catalytic pyrolysis of Karanja seed to produce liquid fuel, *Fuel* 115 (2014) 434–442, <https://doi.org/10.1016/j.fuel.2013.07.053>.
- [24] X. Xu, Simultaneous removal of SO₂ and trace As₂O₃ from flue gas: mechanism, kinetics study, and effect of main gases on arsenic capture 41 (2007) 2894–2900, <https://doi.org/10.1021/es061849a>.
- [25] R.A. Jadhav, L. Fan, Capture of gas-phase arsenic oxide by Lime: Kinetic and mechanistic studies 35 (2001) 794–799, <https://doi.org/10.1021/es001405m>.
- [26] R. Xiao, W. Yang, Influence of temperature on organic structure of biomass pyrolysis products, *Renew. Energy* 50 (2013) 136–141, <https://doi.org/10.1016/j.renene.2012.06.028>.
- [27] S.C. Pereira, L. Maehara, C.M.M. Machado, C.S. Farinas, Physical-chemical-morphological characterization of the whole sugarcane lignocellulosic biomass used for 2G ethanol production by spectroscopy and microscopy techniques, *Renew. Energy* 87 (2016) 607–617, <https://doi.org/10.1016/j.renene.2015.10.054>.
- [28] K.P. Shadangi, K. Mohanty, Kinetic study and thermal analysis of the pyrolysis of non-edible oilseed powders by thermogravimetric and differential scanning calorimetric analysis, *Renew. Energy* 63 (2014) 337–344, <https://doi.org/10.1016/j.renene.2013.09.039>.
- [29] S.R. Tang, Z.M. Zong, L. Zhou, W. Zhao, X.B. Li, Y.L. Peng, R.L. Xie, X.F. Chen, W.T. Gu, X.Y. Wei, Molecular composition of soluble fraction from depolymerized cornstarch powder in supercritical methanol and ethanol, *Renew. Energy* 35 (2010) 946–951, <https://doi.org/10.1016/j.renene.2009.11.037>.
- [30] Y. Yu, Y. Long, H. Wu, Near-complete recovery of sugar monomers from cellulose and lignocellulosic biomass via a two-step process combining mechanochemical hydrolysis and dilute acid hydrolysis, *Energy Fuels* 30 (2016) 1571–1578, <https://doi.org/10.1021/acs.energyfuels.5b02196>.
- [31] O. Onay, O.M. Kockar, Slow, fast and flash pyrolysis of rapeseed, *Renew. Energy* 28 (2003) 2417–2433, [https://doi.org/10.1016/S0960-1481\(03\)00137-X](https://doi.org/10.1016/S0960-1481(03)00137-X).
- [32] F. Wei, J.P. Cao, X.Y. Zhao, J. Ren, J.X. Wang, X. Fan, X.Y. Wei, Nitrogen evolution during fast pyrolysis of sewage sludge under inert and reductive atmospheres, *Energy Fuels* 31 (2017) 7191–7196, <https://doi.org/10.1021/acs.energyfuels.7b00920>.
- [33] H. Egu, Calibrated Quantitative Thermogravimetric Analysis for the Determination of Portlandite and Calcite Content in Hydrated Cementitious Systems, 2017, <https://doi.org/10.1617/s11527-017-1046-2>.
- [34] F. Stalport, P. Coll, M. Cabane, A. Person, F. Raulin, M.J. Vaulay, P. Ausset, C.P. McKay, C. Szopa, J. Zarnecki, Search for past life on Mars: physical and chemical characterization of minerals of biotic and abiotic origin: part 1, *Calcite* 32 (2005) 1–5, <https://doi.org/10.1029/2005GL023403>.
- [35] M. Thiery, G. Villain, P. Dangla, G. Platret, Investigation of the carbonation front shape on cementitious materials: effects of the chemical kinetics 37 (2007) 1047–1058, <https://doi.org/10.1016/j.cemconres.2007.04.002>.
- [36] L. Zeng, X. Hu, N. Gu, B. Fu, C. Qin, Investigation of volatile chemicals and their distributions from pyrolysis of chitin by FT-IR and GC-MS, *J. Anal. Appl. Pyrolysis* 112 (2015) 357–362, <https://doi.org/10.1016/j.jaap.2014.12.026>.
- [37] Q. Liu, S. Wang, Y. Zheng, Z. Luo, K. Cen, Mechanism study of wood lignin pyrolysis by using TG-FTIR analysis, *J. Anal. Appl. Pyrolysis* 82 (2008) 170–177, <https://doi.org/10.1016/j.jaap.2008.03.007>.
- [38] S. Wang, G. Dai, H. Yang, Z. Luo, Lignocellulosic biomass pyrolysis mechanism: a state-of-the-art review, *Prog. Energy Combust. Sci.* 62 (2017) 33–86, <https://doi.org/10.1016/j.pecs.2017.05.004>.
- [39] B.A. Stankiewicz, P.F. van Bergen, I.J. Duncan, J.F. Carter, D.E.G. Briggs, R.P. Evershed, Recognition of chitin and proteins in invertebrate cuticles using analytical pyrolysis/gas chromatography and pyrolysis/gas chromatography/mass spectrometry, *Rapid Commun. Mass Spectrom.* 10 (1996) 1747–1757, [https://doi.org/10.1002/\(SICI\)1097-0231\(199611\)10:14<1747::AID-RCM713>3.0.CO;2-H](https://doi.org/10.1002/(SICI)1097-0231(199611)10:14<1747::AID-RCM713>3.0.CO;2-H).
- [40] G.T. Kjartansson, S. Zivanovic, K. Kristbergsson, J. Weiss, Sonication-assisted extraction of chitin from North Atlantic shrimps (*Pandalus borealis*), *J. Agric. Food Chem.* 54 (2006) 5894–5902, <https://doi.org/10.1021/jf060646w>.
- [41] B. Artur Stankiewicz, M. Mastalerz, C.H.J. Hof, A. Bierstedt, M.B. Flannery, D.E.G. Briggs, R.P. Evershed, Biodegradation of the chitin-protein complex in crustacean cuticle, *Org. Geochem.* 28 (1998) 67–76, [https://doi.org/10.1016/S0146-6380\(97\)00113-7](https://doi.org/10.1016/S0146-6380(97)00113-7).
- [42] S.A. Channiwala, P.P. Parikh, A unified correlation for estimating HHV of solid, liquid and gaseous fuels, *Fuel* 81 (2002) 1051–1063.
- [43] L.J. McKenzie, F.J. Tian, C.Z. Li, Effects of volatile-char interaction on the formation of HCN and NH₃ during the gasification of Victorian brown coal in O₂ at 500 °C, *Fuel* 85 (2006) 2148–2154, <https://doi.org/10.1016/j.fuel.2006.03.024>.
- [44] S. Kajitani, H.L. Tay, S. Zhang, C.Z. Li, Mechanisms and kinetic modelling of steam gasification of brown coal in the presence of volatile-char interactions, *Fuel* 103 (2013) 7–13, <https://doi.org/10.1016/j.fuel.2011.09.059>.